Day 3 — Controller Installation and the Enterprise Console

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Blueprint 2.1 and 4.3

Goal: install the Enterprise Console and a Controller three ways (GUI, CLI, silent), verify and operate it, and use the Administration Console.
Verification: every fact checked against the current Splunk AppDynamics on-prem docs (October 2026).

Block 1 — The install roadmap

```
1. Prepare host  →  2. Install Enterprise Console  →  3. Create platform + add hosts
→  4. Install Controller (Express / Custom / CLI)  →  5. Verify  →  6. Secure (Day 4)
```

Three accounts are set during install:

Account	Purpose
Embedded MySQL root	The Controller's database
Controller root user	Built-in global administrator — used for the Administration Console
Controller administrator	Normal admin login to the Controller UI (default username **admin**)

Usernames and passwords must **not** contain **&** or **!**.

Block 2 — Prepare the Linux host

- Libraries: `libaio`, `numactl` (NUMA hosts), `tzdata` (Ubuntu), ncurses 5 compatibility, `net-tools`.
- User limits: `nofile` 65535 and `nproc` 8192, hard and soft.
- Install user: read/write/execute on the install directory, plus write access to `/etc/.java/.systemprefs`.
- Antivirus: exclude `<controller_home>/db/data`.

Block 3 — Install the Enterprise Console

Installer: `platform-setup-64bit-linux.sh` (Windows: `platform-setup-64bit-windows.exe`). Run with GUI/console prompts, or silently:

```
./platform-setup-64bit-linux.sh -q -varfile ~/response.varfile
```

response.varfile (Linux) — values from the docs:

```
serverHostName=ec01.bank.local
sys.languageId=en
disableEULA=true
sys.installationDir=/opt/appdynamics/platform
platformAdmin.port=9191
platformAdmin.databasePort=3377
platformAdmin.adminPassword=<pwd>
platformAdmin.databasePassword=<pwd>
platformAdmin.databaseRootPassword=<pwd>
platformAdmin.useHttps$Boolean=false
```

- Always use an **absolute** installation path. On AWS, serverHostName = the **public DNS** name.
- The EC has its **own** MySQL on port **3377**; the Controller's database uses **3388**.
- Open the EC at `http://<ec-host>:9191`. Log: `<platform_admin_home>/logs/platform-admin-server.log`.

Block 4 — Install the Controller

	Express Install	Custom Install
Installs	Single-node Controller + **embedded** Events Service on one host	Whatever you choose
Version	Latest only	You pick
HA pair	No	Yes — primary + optional secondary host
Tenancy mode	Default (single)	Choosable
Events Service	Embedded	Separate: 1 node or 3 or more
Use for	Labs, quick start	Anything real

- SSH keys must be **PEM format** (convert newer OpenSSH / Ed25519 keys).
- The EC **can't manage remote Windows hosts** — a Windows Controller needs the EC on the same machine.
- The **Controller install is the longest job**. Default Java is **Azul Zulu** (Azul Prime optional).
- Platform path must contain no existing AppDynamics components. From the CLI, create the platform **first**.

CLI install (run in <EC>/platform-admin):

```
./platform-admin.sh create-platform --name bank --installation-dir /opt/appdynamics/product
./platform-admin.sh add-credential --credential-name ctrlkey --type ssh \
    --user-name appduser --ssh-key-file /home/appduser/.ssh/id_rsa
./platform-admin.sh add-hosts --hosts ctrl01 --credential ctrlkey
./platform-admin.sh submit-job --service controller --job install --args \
    controllerPrimaryHost=ctrl01 controllerAdminUsername=admin \
    controllerAdminPassword=<pwd> controllerRootUserPassword=<rootpwd> \
    mysqlRootPassword=<dbrootpwd> controllerProfile=demo
./platform-admin.sh list-job-parameters --job install --service controller
```

"Install the Controller manually" on the blueprint: today the Controller is always installed through the EC; "manually" means Custom Install or the CLI rather than Express. Prep-guide sample answer — the two Linux installation modes are **silent mode and the GUI installer**.

Block 5 — Verify and operate

Verify (as documented):

1. Open `http://<host>:8090/controller/rest/serverstatus` → `<available>true</available>`.
2. Log in as the **initial Controller administrator** (admin).
3. Optional: `bin/platform-admin.sh check-controller-health`.

Start / stop (standalone Linux) — database first, then app server; stop in reverse:

```
platform-admin.sh start-controller-db --platform-name bank
platform-admin.sh start-controller-appserver --platform-name bank
platform-admin.sh stop-controller-appserver
platform-admin.sh stop-controller-db
```

- **Windows:** services *AppDynamics Controller Application Server* and *AppDynamics Database*. Starting the app server starts the database; stopping the app server does **not** stop the database.
- **HA pair:** `bin/startController.sh` / `bin/stopController.sh` or `service appdcontroller` / `appdcontroller-db`. **Disable auto-failover before restarting the primary** — otherwise the **Watchdog** fails over if the app server is down more than **5 minutes**.
- The EC waits up to **45 minutes** for a start/stop job.

Log	Location
Controller app server	<controller_home>/logs/server.log
Controller database	<controller_home>/db/logs/database.log (check for InnoDB: Error before upgrades)
Enterprise Console	<platform_admin_home>/logs/platform-admin-server.log

Block 6 — Default ports (official table)

Port	Component
8090	Controller app server HTTP
8181	Controller app server HTTPS
4848	Controller app server admin
3388	Controller database (MySQL)
3377	Enterprise Console database
9191	Enterprise Console
8020 / 8021	Reporting Service HTTP / HTTPS
8030	Fleet Management Service
9080 / 9081	Events Service REST API / admin
7001 / 7002	EUM Server HTTP / HTTPS

Block 7 — The Administration Console (4.3)

- URL: `http://<host>:8090/controller/admin.jsp` — log in as **root** with the root user password.
- Already logged in as another user → **Access Denied**. Log out first or use a private window.
- **Controller Settings:** platform-wide properties (EUM and Events Service URLs/keys, metric retention).
- **Account Settings:** accounts, licenses, access keys.
- Single-tenant default account: **customer1**. REST login: `<user>@customer1:<password>`.
- Self-monitoring: `/controller?enableAccounts=true` → account **system**, user **root**.

Property	Data	Default (on-prem)
metrics.min.retention.period	1-minute	**4 hours**
metrics.ten.min.retention.period	10-minute	**48 hours**
metrics.retention.period	1-hour	**365 days**

> Each value must be larger than the one above it. Longer retention grows the database quickly. (SaaS keeps 1-minute data up to 8 days — don't mix them up.)

Paper lab (~40 min)

1. **Response file:** install the EC silently at /opt/appdynamics/platform on ec01.bank.local, GUI over HTTP on port 9191.
2. **CLI commands:** install a **small** profile Controller on remote host ctrl01 with the key /home/appduser/.ssh/appd.pem.
3. **Troubleshoot:** after install, serverstatus doesn't respond — list the first three checks and the log files you'd open.
4. **Admin task:** keep 1-minute data for 8 hours — which property, where, and what risk?

Day 3 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. Which command installs the Enterprise Console silently on Linux?

A. ./platform-setup-64bit-linux.sh -q -varfile ~/response.varfile
B. ./platform-admin.sh install --silent
C. ./controller.sh install -q
D. ./platform-setup-64bit-linux.sh --express
Answer: A — -q plus a response file passed with -varfile; use an absolute install path.

Q2. What does Express Install deploy?

A. An HA pair and a 3-node Events Service cluster
B. A single-node Controller and an embedded Events Service on one host
C. Controller, Events Service and EUM Server
D. A Controller of any version you choose
Answer: B — Latest version only; Custom Install gives choices. The EC never installs the EUM Server.

Q3. Standalone Linux Controller — correct start order?

A. App server, then database
B. Database, then app server
C. Both with start-controller-all
D. Reporting Service, then database
Answer: B — start-controller-db, then start-controller-appserver; stop in reverse.

Q4. Default port of the Enterprise Console's own database?

A. 3388
B. 3377
C. 4848
D. 9191
Answer: B — EC database 3377; Controller database 3388; 4848 app server admin; 9191 EC UI.

Q5. Logged in as admin, a user opens admin.jsp in the same browser and gets Access Denied. Fix?

A. Restart the Controller
B. Log out (or private window), then log in as root
C. Give admin the Account Owner role
D. Enable HTTPS on 8181
Answer: B — The Administration Console needs the root user; log out first.

Q6. Default retention for 1-minute metric data on an on-prem Controller?

A. 4 hours
B. 48 hours
C. 8 days
D. 365 days
Answer: A — metrics.min.retention.period = 4 hours on-prem (8 days is SaaS).

Q7. Which two documented steps verify a new Controller installation?

A. Open /controller/rest/serverstatus and log in as the initial administrator
B. mysql -u root and ping 9191
C. Check license page and restart EC
D. Install a Java agent and wait
Answer: A — Browse to serverstatus, then log in with the initial administrator.

Q8. Before restarting the primary Controller of an HA pair, you must:

A. Stop the secondary's database
B. Disable auto-failover, then re-enable it afterwards
C. Uninstall the HA modules
D. Switch to multi-tenant
Answer: B — Otherwise the Watchdog fails over if the app server is down more than 5 minutes.

Q9. Basic-auth REST username format for a single-tenant on-prem Controller?

A. admin:password
B. admin@customer1:password
C. root@system:password
D. customer1\admin:password
Answer: B — Default account is customer1: <user>@customer1:<password>.

Q10. EC on Linux, Controller wanted on a remote Windows server — result?

A. Works with WinRM
B. Works with a PEM key
C. Not supported: the EC must run on the same Windows machine
D. Works for Large only
Answer: C — The EC doesn't manage remote Windows hosts.

Quick-review cheat sheet

Remember	Value
EC installer	platform-setup-64bit-linux.sh · silent: -q -varfile response.varfile
EC ports	UI 9191 · own database 3377
Accounts at install	MySQL root · Controller root user · Controller admin (default 'admin'); no & or !
Express vs Custom	Express = 1 host, embedded ES, latest only · Custom = versions, HA, separate ES
EC limits	PEM keys · no remote Windows hosts · Controller install is the longest job
Verify	/controller/rest/serverstatus + log in as initial admin · check-controller-health
Start / stop order	DB then app server · stop app server then DB
HA restart	Disable auto-failover first · Watchdog fails over after 5 min
Logs	logs/server.log · db/logs/database.log · platform-admin-server.log
Admin Console	/controller/admin.jsp · root user · private window if Access Denied
Retention defaults (on-prem)	1-min 4 h · 10-min 48 h · 1-hour 365 days
Default account	customer1 → user@customer1:password

Sources

- Silent Installation (Enterprise Console)
- Custom Install
- Install the Controller Using the Command Line

- Admin Accounts Created at Installation
- Start or Stop the Controller
- Platform Component Ports
- Platform Log Files
- Modify Metric Data Retention Periods
- Using the Controller APIs
- Access the Administration Console

Paper lab — model answers

How an implementer would answer each Day 3 paper-lab task, with the rule behind each step.

1. Silent Enterprise Console install on ec01.bank.local

response.varfile (absolute paths; HTTP GUI on 9191):

```
serverHostName=ec01.bank.local
sys.languageId=en
disableEULA=true
sys.installationDir=/opt/appdynamics/platform
platformAdmin.port=9191
platformAdmin.databasePort=3377
platformAdmin.adminPassword=<pwd>
platformAdmin.databasePassword=<pwd>
platformAdmin.databaseRootPassword=<pwd>
platformAdmin.useHttps$Boolean=false
```

Command:

```
./platform-setup-64bit-linux.sh -q -varfile /home/appduser/response.varfile
```

- `platformAdmin.useHttps$Boolean=false` gives the GUI over HTTP; `platformAdmin.port=9191` sets its port.
- 3377 is the EC's own database port — not the Controller's 3388.
- Check: open `http://ec01.bank.local:9191`; if it fails, read `<platform_admin_home>/logs/platform-admin-server.log`.

2. CLI install of a small Controller on remote host ctrl01

Run in <EC>/platform-admin. From the CLI the platform must be created **first**:

```
./platform-admin.sh create-platform --name bank --installation-dir /opt/appdynamics/product
./platform-admin.sh add-credential --credential-name ctrlkey --type ssh \
    --user-name appduser --ssh-key-file /home/appduser/.ssh/appd.pem
./platform-admin.sh add-hosts --hosts ctrl01 --credential ctrlkey
./platform-admin.sh submit-job --platform-name bank --service controller \
    --job install --args \
    controllerPrimaryHost=ctrl01 controllerAdminUsername=admin \
    controllerAdminPassword=<pwd> controllerRootUserPassword=<rootpwd> \
    mysqlRootPassword=<dbrootpwd> controllerProfile=small
```

- `appd.pem` is the **private** key and must be in **PEM format** (convert newer OpenSSH / Ed25519 keys).
- ctrl01 must be **Linux** — the EC does not manage remote Windows hosts.
- Not sure of a parameter? `./platform-admin.sh list-job-parameters --job install --service controller` lists them all.
- Expect the Controller install to be the longest job; watch it on the EC Jobs page.

3. serverstatus doesn't respond after install

First three checks, in order:

1. **Did the install job succeed?** Check the EC Jobs page or `platform-admin-server.log`. A failed job means the Controller was never fully installed.
2. **Are the database and app server running?** Run `bin/platform-admin.sh check-controller-health`; confirm ports with `ss -tlnp | grep -E "3388|8090"`. If down, start the **database first**, then the app server.
3. **Is it reachable?** Test from the Controller itself (`curl http://localhost:8090/controller/rest/serverstatus`). If localhost works but remote

doesn't, it's the firewall or network (open 8090/8181).

Log to open	What it tells you
<platform_admin_home>/logs/platform-admin-server.log	Why an EC job (install/start) failed
<controller_home>/logs/server.log	App server start-up errors
<controller_home>/db/logs/database.log	MySQL errors (e.g. InnoDB: Error)

Also re-check host preparation: user limits (nofile 65535, nproc 8192), libaio, free disk space.

4. Keep 1-minute data for 8 hours

- **Property:** `metrics.min.retention.period` — change from the default 4 hours to 8 hours.
- **Where:** Administration Console (/controller/admin.jsp, log in as root) → **Controller Settings** → edit → **Save**.
- **Rule:** it must stay below `metrics.ten.min.retention.period` (default 48 hours) — 8 h is fine.
- **Risk to mention:** longer retention makes the database grow quickly — confirm disk capacity first. If 1-minute + 10-minute retention exceed `caches.retention.period`, the UI slows because data comes from the database instead of cache.
- Note: changing retention does not change the time-range options shown in the Controller UI.